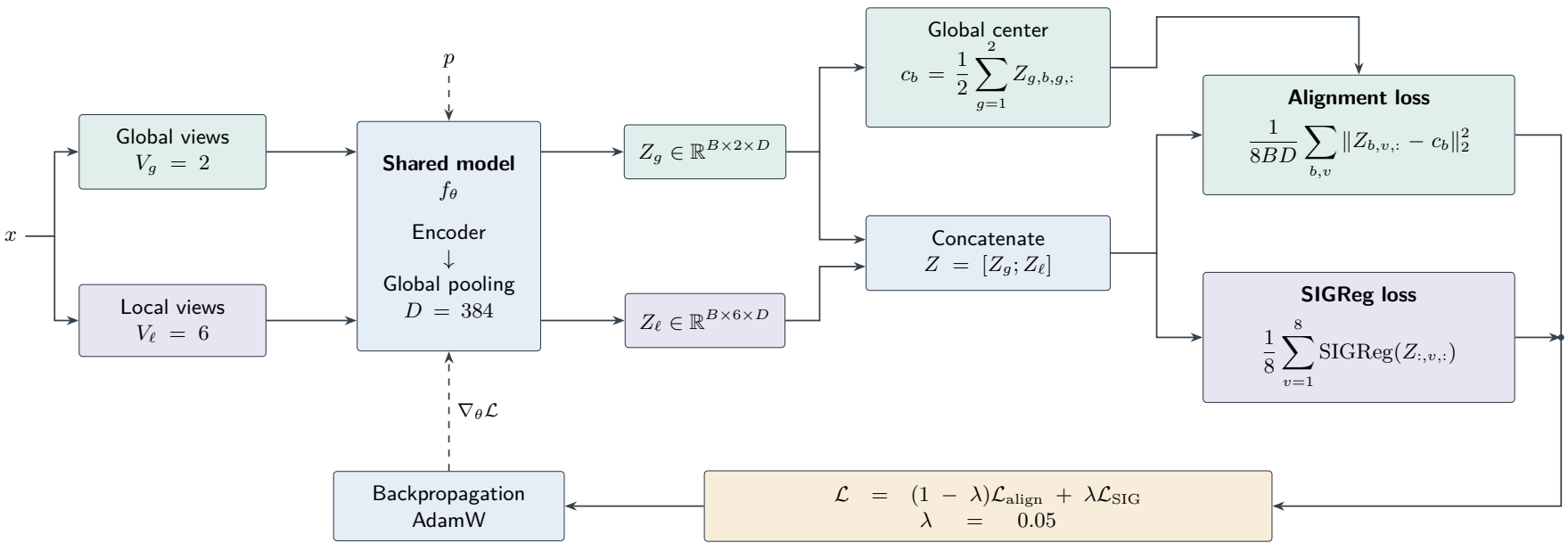

x
Global views
$V_g = 2$
Local views
$V_\ell = 6$
p
Shared model
f_θ
Encoder
↓
Global pooling
$D = 384$
$Z_g \in \mathbb{R}^{B\times 2\times D}$
$Z_\ell \in \mathbb{R}^{B\times 6\times D}$
Global center
$c_b = \frac{1}{2}\sum_{g=1}^{2} Z_{g,b,g,:}$
Concatenate
$Z = [Z_g; Z_\ell]$
Alignment loss
$\frac{1}{8BD}\sum_{b,v} \|Z_{b,v,:} - c_b\|_2^2$
SIGReg loss
$\frac{1}{8}\sum_{v=1}^{8} \mathrm{SIGReg}(Z_{:,v,:})$
$\nabla_\theta \mathcal{L}$
Backpropagation
AdamW
$\mathcal{L} = (1-\lambda)\mathcal{L}_{\text{align}} + \lambda\mathcal{L}_{\text{SIG}}$
$\lambda = 0.05$